

1. Use the definition of a logarithmic function to write the logarithmic equation as an equation involving an exponent.

$$\log_4 J = K$$

The equivalent exponential equation is .

(Type an equation. Do not simplify.)

2. Write the exponential equation $x = 2^y$ in logarithmic form.

The equivalent logarithmic equation is .

- *3. Express as a single logarithm.

$$\log_c(81) + \log_c(7)$$

$$\log_c(81) + \log_c(7) = \text{}$$

- *4. Express as a product.

$$\log C^{14}$$

$$\log C^{14} = \text{}$$

5. Write as a single logarithm. Assume that variables represent positive numbers.

$$9 \log_5 x + 4 \log_5 z$$

$$9 \log_5 x + 4 \log_5 z = \text{}$$

6. Write the logarithmic equation $y = \ln(5x)$ in exponential form.

The equivalent exponential form is .

7. Change the exponential statement to an equivalent statement involving a logarithm.

$$216 = 6^3$$

The equivalent logarithmic statement is . (Type an equation.)

8. The Richter scale measurement for a deep sea earthquake of 2010 was 7.3. Find the intensity of this earthquake as a multiple of I_0 .

The intensity of the earthquake was approximately I_0 .

(Round to the nearest whole number as needed.)

9. To simplify the measurement of the acidity or basicity of a solution, the pH (hydrogen potential) measure was developed. The pH is given by the formula $\text{pH} = -\log[H^+]$, where $[H^+]$ is the concentration of hydrogen ions in moles per liter in the solution. Find the pH value of a solution for which $[H^+] = 0.0000694$.

The pH value of a solution is approximately . (Round to the nearest tenth as needed.)